

University of Rajshahi · Dept. of Computer Science & Engineering

CSE4121

Object Oriented Design and Design Patterns

Final Exam Prediction & Two Model Question Papers

B.Sc. (Engg.) Part-IV Odd Semester

Built from: 2021, 2023, 2024 final papers · CT-1 · CT-2 · Part-A slides (131 pages) · Section-B pattern notes (11 patterns) · handwritten solutions.

How to use this set of three PDFs

PDF 1 (this file) tells you *what* is likely to come and gives two full model papers.

PDF 2 (Bangla theory) explains *every* topic in Bangla with UML and code.

PDF 3 (English Q&A) gives exam-ready written answers for all 58 questions — every past-paper and class-test question plus every model-paper question, with UML diagrams and Java code that has been compiled and run.

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CHAPTER 1

Exam Pattern and Source Analysis

1.1 The exam at a glance

Course	CSE4121 — Object Oriented Design and Design Patterns
Time / Full marks	2 hours / 35 marks
Rule	Answer four questions taking two from each section (6 questions printed)
Marks per question	8.75 (usually split 3 + 4.75 + 1, or 4.5 + 4.25, or 5 + 3.75)
Section A (Q1–Q3)	Design principles: OO modelling, CRC cards, decomposition, inheritance, coupling/cohesion, code quality, UML diagrams
Section B (Q4–Q6)	Design patterns: definitions, categories, Singleton, Adapter, Observer, Chain of Responsibility, Facade, ...

Time budget that works

2 hours for 4 questions = **28 minutes per question**. Spend the first 4 minutes choosing questions. For a 4.5-mark design-pattern question write: pattern name + one-line intent (1 min), UML class diagram (6–7 min), Java code of the key classes (10 min), 2–3 lines on why it fits (2 min).

1.2 What was analysed

Source	What it contributed
Final 2024	First paper using the Dhaka Urban Management System (DUMS) and Agricultural Automation System (AAS) scenarios; introduced code smells and SRP.
Final 2023	ATM CRC cards, Whale inheritance problem, Sensor coupling/cohesion, pattern categories, logging Singleton, bank cheque Chain of Responsibility.
Final 2021	OO modelling, CRC, TV-remote sequence diagram, SmartPhone coupling/cohesion + separation of concerns, Database Singleton, Adapter, MVC, Observer, Facade.
CT-1 (20 marks)	UML necessity vs IDE code generation; DUMS association/aggregation/composition; AAS inheritance redesign.
CT-2 (15 marks)	DUMS complaint notification (Observer); university approval chain (Chain of Responsibility); IoT soil sensor (Adapter).
Part-A slides	3 modules: OOAD & modelling, four design principles + UML class diagrams, coupling/cohesion/SoC/information hiding/conceptual integrity + sequence & state diagrams + model checking.
Section-B notes	GoF introduction + Factory, Abstract Factory, Singleton, Builder, Adapter, Bridge, Filter, Composite, Command, Mediator, Observer.
Handwritten solve	Solutions for some 2021/2023 questions — two flaws found (see below).
Old CSE2211/2121 papers	Different (Java programming) course — ignored for prediction.

Two problems in the handwritten solution file — do not copy them

1. “Chain of Responsibility, MVC and Decorator are not in syllabus.” Chain of Responsibility was asked in the 2023 final (bank cheque) *and* in CT-2 (Teacher → Chairman → Dean → VC). MVC was asked in 2021. Prepare CoR seriously; MVC at least at a descriptive level.
2. The Database Singleton does not compile. It calls new Database() but the only constructor takes three arguments. The corrected, compiled version is in PDF 3, Q36.

CHAPTER 2

Topic Frequency Heat-Map

2.1 Every topic, every paper

Topic	2021	2023	2024	CT	Source	Prediction
SECTION A — Object-Oriented Design						
Association / aggregation / composition + UML	2(a)	2(a)	2(a)	CT-1 Q2	Slides	Certain
CRC cards for a scenario	1(c)	1(b)	1(b)	—	Slides	Very high
Software design vs architecture	1(b)	1(c)	1(c)	—	Slides	Very high
Objects for real-world entities / OO modelling	1(a)	1(a)	1(a)	—	Slides	Very high
Inheritance design or misuse	—	2(b)	2(c)	CT-1 Q3	Slides	Very high
Coupling & cohesion of a flag-based class	3(b)	3(a)	—	—	Slides	Very high
Separation of concerns / redesign + UML	3(c)	3(b)	—	—	Slides	High
Conceptual integrity	3(a)	—	5(b)	—	Slides	High
UML sequence diagram	2(b)	—	5(a)	—	Slides	High
Encapsulation, access modifiers, black box	—	—	2(b)	—	Slides	High
Code smells, SRP, comments as smell	—	—	3(a–c)	—	Paper only	High
UML vs IDE / conceptual vs technical design	—	—	—	CT-1 Q1	Slides	Medium
UML state diagram, model checking	—	—	—	—	Slides	Medium (new)
Entity–control–boundary, trade-offs, quality	—	—	—	—	Slides	Medium (new)
Information hiding, Law of Demeter, OCP	—	—	—	—	Slides	Medium (new)
SECTION B — Design Patterns						
What are design patterns / why learn them	4(a)	4(a), 5(a)	4(a)	—	Notes	Certain
Singleton (convert a class / shared service)	4(b)	5(c)	4(b)	—	Notes	Certain
Adapter (describe + UML, or scenario + code)	4(c)	6(a)	4(c), 6(b)	CT-2 Q3	Notes	Certain
Observer	5(b)	—	6(a)	CT-2 Q1	Notes	Very high
Chain of Responsibility	—	6(b)	—	CT-2 Q2	Paper only	Very high
Pattern categories	—	4(b,c), 5(b)	—	—	Notes	High
Factory / Abstract Factory	—	—	—	—	Notes	High (new)
Builder, Composite, Command	—	—	—	—	Notes	High (new)
Facade	6(a–c)	—	—	—	Paper only	Medium
Bridge, Filter, Mediator	—	—	—	—	Notes	Medium (new)
MVC	5(a)	—	—	—	Paper only	Medium
Decorator	—	—	—	—	Solve note	Low

Cells give the question number where the topic appeared. *Slides* = Part-A slides, *Notes* = Section-B pattern notes, *Paper only* = asked in a paper or CT but not covered by the notes.

2.2 What the pattern of past papers tells us

1. **Section A has a fixed skeleton.** Q1 is always “modelling” (objects + CRC + design vs architecture), Q2 is always “decomposition + one inheritance question”, and Q3 is a code-reading question (coupling/cohesion in 2021 and 2023, code smells in 2024). Expect the same skeleton with a DUMS

or AAS scenario.

2. **The scenario is now DUMS/AAS.** The 2024 paper, CT-1 and CT-2 all used the Dhaka Urban Management System and the Agricultural Automation System from the slides. Practise every principle on those two systems — not on ATM or generic animals.
3. **Coupling/cohesion skipped 2024, so it is due.** The last slide of Module 3 contains an unsolved exercise (`ServiceManager.processRequest(int flag)` in DUMS). It is the single most likely Section A Q3.
4. **Singleton and Adapter appeared in every final.** “What are design patterns” also appeared in every final. These three alone can secure one Section B question.
5. **CT-2 previews Section B.** Observer, Chain of Responsibility and Adapter were all in CT-2 with DUMS/AAS scenarios — expect at least two of them again.
6. **Eight patterns from the notes were never asked.** Factory, Abstract Factory, Builder, Bridge, Filter, Composite, Command and Mediator were taught but never examined, so one “new” question is likely to come from this group.

2.3 Priority tiers

Tier	Topics	Q&A (PDF 3)
Tier 1 — must know	Decomposition + UML (DUMS); CRC cards; design vs architecture; inheritance design/misuse; coupling & cohesion with a flag class; design patterns definition; Singleton conversion; Adapter; Observer; Chain of Responsibility	Q4–6, Q13–18, Q20–23, Q33, Q35–37, Q41–43, Q49–54
Tier 2 — very likely	Conceptual integrity; sequence diagram; encapsulation/access modifiers; SoC; code smells & SRP; pattern categories; Factory; Builder; Composite; Command; Facade	Q12, Q24–30, Q34, Q38, Q40, Q44, Q45, Q55
Tier 3 — possible	State diagram; entity/control/boundary; trade-offs; Abstract Factory; Bridge; Filter; Mediator; MVC; OCP; Law of Demeter	Q7–11, Q19, Q27, Q28, Q31, Q32, Q39, Q46, Q47, Q56, Q57

Model Question Paper — Set 1

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. (Engg.) Part-IV Odd Semester Examination — **MODEL PAPER (SET 1)**

Course: CSE4121 (Object Oriented Design and Design Patterns)

Time: 2 Hours

Full Marks: 35

Answer any FOUR questions taking TWO from each section. Figures in the right margin indicate full marks.

SECTION A

1. (a) “Good design begins before a single line of code is written.” Explain this statement. Differentiate between *conceptual design* and *technical design* with reference to the waste-collection module of the Dhaka Urban Management System (DUMS). 3
- (b) In an Agricultural Automation System (AAS), soil-moisture sensors in every field send readings every 10 minutes. When the moisture of a field falls below its threshold, the irrigation controller checks the weather forecast; if no rain is expected, it opens the valve of that field and informs the farmer through the mobile app. Using CRC cards, identify the classes of this scenario with their responsibilities and collaborators. 4.75
- (c) Differentiate between software design and software architecture. 1
2. (a) In DUMS, a city corporation is divided into collection zones. Each zone contains many smart garbage bins, and a bin can be moved to another zone. Every bin has a built-in fill-level sensor that cannot exist without the bin. Garbage trucks collect waste from bins. Identify the association, aggregation and composition relationships, draw a UML class diagram with multiplicities, and write the Java skeleton that shows each relationship. 4.75
- (b) List the UML visibility symbols and the equivalent Java access modifiers. 1
- (c) Consider the following AAS code. Identify the design problem and redesign the class hierarchy so that it does not happen. 3

```
public class FarmingMachine {
    protected String machineId;
    protected double fuelLevel;
    public void start() { /* ... */ }
    public void stop() { /* ... */ }
    public void refuel(double litres) { fuelLevel += litres; }
}

public class SolarIrrigationPump extends FarmingMachine {
    @Override
    public void refuel(double litres) {
        throw new UnsupportedOperationException("Solar pump has no fuel tank");
    }
}
```

3. The ServiceManager class of DUMS handles two different municipal services depending on a control flag:

```
public class ServiceManager {
    public void processRequest(int controlFlag) {
        switch (controlFlag) {
            case 0: this.handleWasteCollection(); break;
            case 1: this.handleRoadMaintenance(); break;
            default: throw new UnknownControlFlagException();
        }
    }
    private void handleWasteCollection() { /* find truck, assign route ... */ }
    private void handleRoadMaintenance() { /* find crew, order material ... */ }
}
```

- (a) Evaluate the design in terms of coupling and cohesion. Justify your answer. 4.5
- (b) Suggest a better design with a UML class diagram that meets acceptable coupling and cohesion. Show how a new *water-logging relief* service can be added to your design. 4.25

SECTION B

4. (a) What are software design patterns? Explain two principles of reusable object-oriented design proposed by the Gang of Four. 2.5
- (b) Classify the GoF design patterns into three categories with two examples of each. 2.5
- (c) Several DUMS modules (housing, garbage, traffic) create their own object of the following class, and every object opens a new costly connection: 3.75

```
class SmsGateway {
    private String connectionId;
    public SmsGateway() {
        connectionId = openConnection("sms.dums.gov.bd", 8443); // slow, costly
    }
    public void send(String phone, String text) {
        System.out.println "[" + connectionId + "] to " + phone + ": " + text);
    }
}
```

Convert the class into a thread-safe, lazily initialised Singleton and show how the modules use it.

5. (a) A citizen logs into the DUMS complaint portal with NID and password and submits a complaint about uncollected garbage. The portal verifies the citizen through an authentication service, creates a complaint, and asks the collection scheduler for a truck. If a truck is available, the scheduler assigns the complaint to the truck (the driver is notified) and marks it ASSIGNED; otherwise it queues the complaint and marks it QUEUED. The citizen receives a ticket number and status. Draw a UML sequence diagram for this interaction, using an alt fragment. 5
- (b) A DUMS complaint is *submitted*, then *reviewed* by an officer. An invalid or duplicate complaint is *rejected*; a valid one is *assigned* to a field team, goes *in progress* and becomes *resolved*. The citizen can confirm (complaint *closed*) or reopen it within 7 days; otherwise it is auto-closed after 7 days. Draw a UML state diagram for a Complaint object. 3.75

6. (a) In AAS, a WeatherStation publishes temperature, humidity and rainfall. The irrigation controller, harvest planner and farmer dashboard must react whenever a new reading arrives, and new modules (e.g. a pest-risk analyser) will be added later without changing the station. Which design pattern is suitable? Explain with a UML class diagram and Java code. 4.5
- (b) DUMS escalates complaints by severity: 1–3 are resolved by the Ward Council-lor, 4–6 by the Zone Executive Officer, 7–8 by the Chief Executive Officer and 9–10 by the Mayor. The sender should not need to know who finally resolves the complaint. Implement this using an appropriate design pattern. 4.25

CHAPTER 4

Model Question Paper — Set 2

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. (Engg.) Part-IV Odd Semester Examination — **MODEL PAPER (SET 2)**

Course: CSE4121 (Object Oriented Design and Design Patterns)

Time: 2 Hours

Full Marks: 35

Answer any FOUR questions taking TWO from each section. Figures in the right margin indicate full marks.

SECTION A

1. (a) What is object-oriented modelling? Classify the following AAS classes as entity, control or boundary objects, with a reason for each: Field, Crop, SoilSensorReading, IrrigationScheduler, HarvestPlanner, FarmerMobileScreen, WeatherAPIGateway. 3
- (b) The client of DUMS wants every citizen to see the real-time location of all garbage trucks, while the development team has limited time, budget and server capacity. Discuss the trade-offs between client needs and developer scope, with reference to software quality attributes. 4.75
- (c) Differentiate between functional and non-functional requirements with one DUMS example of each. 1
2. (a) Design a TrafficLight class for DUMS applying abstraction and encapsulation: the signal state must never take an invalid value, the green duration must stay between 10 and 120 seconds, and the way traffic flow is detected must be hidden from other classes. Draw the UML class diagram and write Java code. 3.5
- (b) An AAS smart drone is a farming machine that can both fly over fields and spray pesticide. Java does not allow a class to extend more than one class. Show with a UML diagram and Java code how interface inheritance solves this. 3
- (c) Compare aggregation and composition with suitable DUMS examples. 2.25

3. (a) Consider the following code of a DUMS toll plaza: 3.5

```
public class TollCalculator {
    public double calculateToll(String vehicleType, int axles) {
        switch (vehicleType) {
            case "CAR": return 750;
            case "BUS": return 1500;
            case "TRUCK": return 1000 + axles * 500;
            case "MOTORCYCLE": return 100;
            default: throw new IllegalArgumentException("Unknown vehicle");
        }
    }
}
```

Identify the code smell. Refactor it with polymorphism so that a new vehicle type can be added without modifying existing code, and name the principle you applied.

- (b) Identify the design problem in the following code and refactor it. Which principle is violated? 3.5

```
public class ComplaintService {
    public void escalate(Complaint c) {
        c.getCitizen().getAddress().getZone().getOfficer().assign(c);
    }
}
```

- (c) Differentiate between information hiding and encapsulation. 1.75

SECTION B

4. (a) DUMS notifies citizens by SMS, e-mail and push notification. Currently every module creates the concrete notification objects with new. Apply the Factory pattern with a UML diagram and Java code, and show how a WhatsApp channel can be added later. 4.5
- (b) A HousingApplication in DUMS has two mandatory fields (applicant name, NID) and five optional fields (family size, monthly income, preferred area, freedom-fighter quota, disability quota), which has produced many overloaded constructors. Apply the Builder pattern with Java code. 4.25
5. (a) The DUMS billing portal pays bills through PaymentProcessor.pay(String billId, double amountTk): boolean. A third-party MobileWalletSDK offers only sendMoney(String walletNo, long amountInPaisa, String reference): String and cannot be modified. Apply the Adapter pattern with UML and code, and distinguish between an object adapter and a class adapter. 4.5
- (b) Registering a new resident in DUMS requires calls to HousingService, UtilityBillingService, GarbageService and NotificationService in the correct order, and the citizen-portal developers find this complex. Which design pattern simplifies this? Explain with a UML diagram and code. 4.25
6. (a) A DUMS city is divided into zones, zones into wards, and wards contain garbage bins. The authority needs the total waste at any level, and the report code must treat a single bin and a whole zone in the same way. Which design pattern is suitable? Explain with a UML diagram and Java code. 4.5
- (b) The AAS farmer app has buttons to start irrigation, stop irrigation and spray pesticide. Commands must be queueable for a night schedule and the last command must be undoable. Apply an appropriate design pattern. 4.25

CHAPTER 5

Answer Map and Revision Plan

5.1 Where each model-paper answer is

All answers are written in full in **PDF 3 (English Q&A)**, and the theory behind them is in **PDF 2 (Bangla)**.

Set 1			Set 2		
Q	Topic	PDF 3	Q	Topic	PDF 3
1(a)	Design before coding	Q1	1(a)	OO modelling, E/C/B	Q7
1(b)	CRC cards (AAS)	Q5	1(b),(c)	Trade-offs, requirements	Q8
1(c)	Design vs architecture	Q6	2(a)	TrafficLight	Q11
2(a)	DUMS decomposition	Q14	2(b)	Drone interfaces	Q19
2(b)	Visibility symbols	Q12	2(c)	Aggregation vs composition	Q15
2(c)	Solar pump refuel	Q18	3(a)	Toll switch, OCP	Q27
3	ServiceManager	Q21	3(b)	Message chain, LoD	Q28
4(a)	Patterns, GoF principles	Q33	3(c)	Info hiding vs encapsulation	Q12
4(b)	Categories	Q34	4(a)	Factory	Q38
4(c)	SmsGateway Singleton	Q37	4(b)	Builder	Q40
5(a)	Sequence diagram	Q30	5(a)	Payment Adapter	Q43
5(b)	State diagram	Q31	5(b)	Facade	Q44
6(a)	Observer (weather)	Q51	6(a)	Composite	Q45
6(b)	CoR (escalation)	Q54	6(b)	Command	Q55

5.2 Suggested 4-day revision plan

Day	Work	PDF 3 questions
Day 1	Theory Chapters 1–4 (Bangla PDF). Write CRC cards for DUMS login scenario and AAS irrigation. Draw all three decomposition diagrams from memory.	Q1–Q10, Q13–Q15
Day 2	Inheritance problems (Whale, solar pump, drone), coupling/cohesion (Sensor, SmartPhone, ServiceManager), code smells, SRP, conceptual integrity, TV sequence diagram.	Q16–Q29
Day 3	Pattern foundations + Singleton (write all three versions), Adapter (media player, soil sensor, wallet), Observer (Facebook, complaint, weather), CoR (bank, university, escalation).	Q33–Q37, Q41–Q43, Q49–Q54
Day 4	Factory, Abstract Factory, Builder, Composite, Command, Facade, Bridge, Filter, Mediator, MVC. Then sit Model Set 1 in 2 hours; mark it with PDF 3.	Q38–Q40, Q44–Q48, Q55–Q58

Writing a design-pattern answer that gets full marks

1. Name the pattern and its category (e.g. “Observer — behavioural”).
2. Map the scenario to the pattern roles (Subject = WeatherStation, Observer = IrrigationController ...).
3. Draw a UML class diagram: *interface/abstract class on top, realisation arrows, the key association*.
4. Write the Java code of the interface, one or two concrete classes, and a short client main.
5. Finish with 2–3 advantages tied to the scenario (loose coupling, open/closed, easy to add a

new module).